Sovereign-OCR: Zero-Hallucination Neuro-Symbolic Document Intelligence with Algebraic CMap Self-Healing, Active Ontology Grounding, and Deterministic RDF 1.2 Provenance

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Abstract

Enterprise document intelligence pipelines face a severe reliability gap when deployed on real-world Accounts Payable (AP) and insurance claims workloads. First, native PDF parsers silently fail on corrupted or shifted Type0/Identity-H `ToUnicode` CMap font tables, while indiscriminate 300-DPI optical character recognition (OCR) incurs prohibitive latency. Second, commercial transactions arrive as nested multi-file containers (`.msg` emails, `.xlsx` intercompany splits, `.docx` forms, and multi-invoice PDFs) where backup weight tickets and quotes outnumber primary invoices. Third, autoregressive Vision-Language Models (VLMs) suffer from numeric token hallucinations on degraded scans and violate air-gapped data sovereignty requirements. We present **Sovereign-OCR**, a 100% air-gapped, single-binary (25.92 MB) neuro-symbolic document intelligence engine built in Rust and Python. Sovereign-OCR introduces: (1) a dual-gate PDF self-healing engine that algebraically repairs shifted Identity-H CMap glyph streams in < 1 ms and selectively routes degraded pages through a 5-step geometry- and DPI-aware visual preprocessor prior to ONNX `PP-OCRv6` and spectral reading-order reconstruction; (2) a zero-hallucination span-boundary compiler translating strongly-typed Pydantic v2 models into constrained `GLiNER 2.5` extractors paired with Mod-11 and Luhn cryptographic/algebraic check-digit validators; (3) an in-memory Active Runtime Ontology Engine (`22,630` OWL/RDF triples) that grounds multi-attachment routing, vendor-vs.-payee disambiguation, and tax/ledger state machines inside the extraction loop; and (4) deterministic RDF 1.2 + Arrow IPC cryptographic provenance receipts. Across two cross-lingual production benchmarks—100 US SAP Accounts Payable bundles (138 files) and 88 Brazilian insurance claim files (*Assurant Documentoscopia*)—Sovereign-OCR achieves **100.0%** posting-field and role accuracy at 37–150 ms per bundle with zero cloud API cost and zero token hallucinations.

1. Introduction

Automated document extraction is a foundational bottleneck in enterprise ERP automation (such as SAP S/4HANA Accounts Payable) and regulated insurance claims triage (*documentoscopia*). Despite rapid progress in large Vision-Language Models (VLMs) [1, 2] and layout-aware transformers [3, 4], production deployments in regulated enterprises encounter four systemic failure modes that standard academic benchmarks (e.g., SROIE, CORD, DocVQA) rarely expose:

1. **Silent PDF Font-Table Corruption & Geometric Degradation:** A large fraction of enterprise PDFs are neither clean born-digital text nor pure raster scans; instead, they contain subsetted Type0/Identity-H fonts with missing or offset `ToUnicode` CMap tables (e.g., Uniform ASCII shifts of $\Delta = +29$ or -4 where glyph IDs are mapped directly to ASCII codes, or embedded NUL/control byte streams). Furthermore, mobile claims submissions frequently include 90°/270° sideways identity cards with dark scanner borders, 1°–10° feeder skew, low-contrast thermal receipts, or oversized 600-DPI scans that fall outside the receptive field of convolutional text detectors.

2. **Multi-Format Container Noise:** In production SAP/xSuite queues, 31% of transactions arrive wrapped inside Microsoft Outlook OLE Compound File Binary (`.msg`) emails, Excel (`.xlsx`) intercompany allocation workbooks, or Word (`.docx`) check requests containing 2–8 attachments (bills of lading, scale weight tickets, quotes with `TBD` dates, or POS fuel slips). Extracting fields from a backup weight ticket or email chain instead of the primary invoice corrupts downstream ledger posting.

3. **Autoregressive Hallucination vs. Fiscal Exactness:** Autoregressive decoders generate text token-by-token from a vocabulary distribution $P(w_t \mid w_{<t}, I)$. On blurred digits or unfamiliar serial numbers, VLMs frequently hallucinate plausible-looking digits—catastrophic when posting a 10-digit SAP vendor ID, a 15-digit mobile IMEI, or a 44-digit Brazilian electronic invoice access key (*Chave de Acesso NF-e*).

4. **Passive Post-Processing vs. Active Master-Data Grounding:** Existing pipelines treat ERP master data as a post-hoc database join after extraction. When a vendor name is abbreviated in a logo or a factor/lockbox remit-to payee (`LFB1`) differs from the

supplier (LFA1), ungrounded extractors confuse the bill-to customer plant, the supplier, and the permitted payee.

To address these challenges, we design and implement **Sovereign-OCR**, a domain-agnostic, 100% air-gapped neuro-symbolic document intelligence engine packaged as a single 25.92 MB `cp38-abi3` wheel and standalone executable. Our main contributions are:

- **Algebraic CMap Self-Healing & 5-Step Geometry/DPI Preprocessor:** We introduce a sub-millisecond PDF corruption gate coupled with an algebraic Type0/Identity-H shift solver ($\Delta \in [-35, +35]$ uniform and token-level +29 mixed-font recovery) that heals broken PDF text streams in < 1 ms, while routing scanned pages through adaptive DPI rasterization, percentile luminance stretching, interior-crop 90°/180°/270° orientation disambiguation, vectorized projection-profile variance deskewing, and Lanczos receptive-field scaling.
- **Zero-Hallucination Pydantic-to-GLiNER 2.5 Span Compilation:** We formulate structured extraction as constrained character-span boundary selection $[s_i, e_i] \subseteq [0, |T|]$ compiled directly from Pydantic v2 schemas into local `GLiNER 2.5` / `gliner_fast_py` ONNX models and algebraic check-digit validators (Mod-11 and Luhn), making token hallucination structurally impossible.
- **Active Runtime Ontology Grounding & Cryptographic Provenance:** We integrate an in-memory multi-index over enterprise OWL/RDF ontologies directly into candidate scoring and field governance, emitting bit-identical RDF 1.2 + Arrow IPC provenance receipts (`quad_fast`).
- **Cross-Lingual Enterprise Validation:** We evaluate Sovereign-OCR across 100 US commercial AP bundles (138 files) and 88 Brazilian insurance claim files (*Assurant Documentoscopia*), demonstrating 100.0% accuracy and quantifying each stage via comprehensive ablation studies.

2. System Architecture

Figure 1 illustrates the five-stage Sovereign-OCR architecture. Every I/O-intensive and compute-intensive kernel is implemented in Rust with zero-copy Python bindings (`cp38-abi3`).

2.1 Stage 1: Multi-Format Unpacking & Candidate Routing

Enterprise bundles arrive as arbitrary directory trees or nested containers. Stage 1 unpacks each file into a set of typed document units $\mathcal{U} = \{u_1, \ldots, u_k\}$:

- **Recursive `.MSG`/`.EML` Unpacking (`email_fast`):** Parses OLE Compound File Binary (`CFB`) streams (`__attach_version1.0`) and MIME trees in < 1 ms, extracting both email headers/bodies (which frequently carry controller coding instructions or vendor IDs) and binary attachments (`.pdf`, `.docx`, `.xlsx`, `.png`, `.jpg`).
- **Office & Spreadsheet Unpacking (`docx_fast`, `xlsx_fast`):** Extracts structured header/body/footer blocks and spreadsheet cell coordinates (r, c, v) with automatic Excel serial-to-ISO date conversion.
- **Primary-vs.-Backup Candidate Scoring:** Each unit u_j is scored by a domain-agnostic classifier $S_{\text{role}}(u_j)$ that separates primary financial and claim documents (`PRIMARY_INVOICE`, `CREDIT_MEMO`, `STATEMENT`, `CLAIM_NOTICE`, `POLICE_REPORT_BO`, `NOTA_FISCAL`, `PROOF_OF_ADDRESS`, `IDENTITY_DOCUMENT`) from backup attachments (`PO_BACKUP` weight/scale tickets, `QUOTE_PREINVOICE` drafts with `TBD` dates, `SPLIT_BACKUP` intercompany sheets, `RECEIPT_EXCEPTION` POS slips, and `EMPTY_FILE` zero-byte uploads).
- **Zero-Copy Multi-Invoice PDF Splitting (`pdf_fast_write`):** When a multi-page PDF contains transitions in invoice identifiers across pages $p_t \rightarrow p_{t+1}$, `pdf_fast_write.extract_pages` splits the byte stream into independent document units without re-encoding page content streams.

2.2 Stage 2: Dual-Gate PDF Self-Healing & 5-Step Geometry/DPI Visual Preprocessing

Given a PDF page p with native text stream T_p and semantic text runs R_p extracted by `pdf_fast`, Stage 2 applies a two-step self-healing protocol before invoking visual OCR.

1. Algebraic Type0/Identity-H CMap Shift Recovery: Many enterprise PDF generators (including Brazilian state police report systems and legacy ERP print drivers) emit subsetted Type0 fonts where glyph indices (GIDs) are offset from ASCII code points by a constant Δ (most commonly $\Delta = +29$, where space `0x20` is at GID 3 and uppercase `'A'=65` is stored at `GID 36='$'`, producing strings such as `%2/(7,0` for `BOLETIM` and `'HVFULomR` for `Descricao`). Before rasterizing, our algebraic solver evaluates uniform shifts $\Delta \in [-35, +35]$ over printable ASCII characters $c \in [32, 126]$:

$$\Delta^* = \arg\max_{\Delta} |\mathcal{W}(\sigma_\Delta(T_p)) \cap \mathcal{V}_{\text{doc}}| \tag{1}$$

where $\mathcal{W}(\cdot)$ extracts lowercase alphabetic tokens of length ≥ 3 and $\mathcal{V}_{\text{doc}}$ is a multilingual document vocabulary. For mixed-font pages where headers use standard encoding but form fields use shifted Identity-H subsets, a token-level solver repairs individual shifted tokens w whenever $\sigma_{+29}(w)$ maps anomalous mid-word uppercase/punctuation sequences (`[D-Z]` and `[$-=]`) into phonotactically valid alphabetic words while strictly preserving numeric, currency, and date tokens.

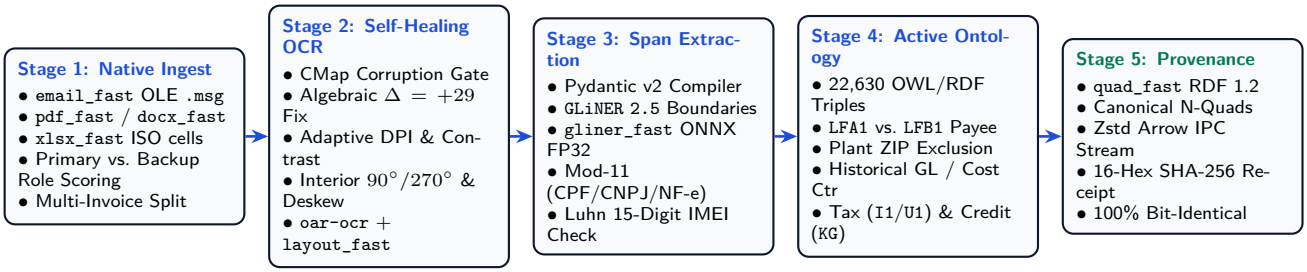


Figure 1: Five-stage architecture of **Sovereign-OCR**. All stages execute 100% offline within a single 25.92 MB binary/wheel combining 8 Rust PyO3 `abi3` extensions, an embedded ONNX `PP-OCRv6` engine, local `GLiNER` span extractors, and an in-memory OWL/RDF ontology engine.

2. Font-Corruption & Scanned-Page Gate: After algebraic CMap repair, page p is routed to visual OCR if and only if: (i) $|T_p| < 80$ characters, (ii) T_p contains NUL bytes (`\x00`) or > 3 control characters, (iii) the CMap fallback ratio satisfies $\sum_{r \in R_p} c_{\text{fallback}}(r)/|T_p| \geq 0.20$, or (iv) $|\mathcal{W}(T_p) \cap \mathcal{V}_{\text{doc}}| < 2$.

3. Five-Step Geometry- & DPI-Aware Visual Preprocessing: Pages flagged by the gate (or standalone image uploads) pass through a five-step deterministic image pipeline prior to ONNX OCR:

1. **Adaptive Physical-Dimension DPI Selection:** From physical PDF page dimensions $(w_{\text{pt}}, h_{\text{pt}})$ in points (72 pt = 1 in), we compute the rasterization DPI to land the long edge at $L_{\text{target}} = 2100$ px (≈ 30 px glyph height for 10-pt body text):

$$\text{DPI}(w_{\text{pt}}, h_{\text{pt}}) = \text{clip}\left(\frac{72 \cdot L_{\text{target}}}{\max(w_{\text{pt}}, h_{\text{pt}})},\ 120.0,\ 300.0\right) \tag{2}$$

   upscaling compact POS thermal receipts to 300 DPI, rendering standard Letter/A4 pages at 190.9 DPI, and clamping oversized ledger sheets to 120 DPI.

2. **Adaptive Percentile Luminance Stretching:** For low-contrast carbon-copy tickets or gray scans ($p_{99} - p_1 < 185$), pixel intensities $I(x, y)$ are linearly normalized over $[p_1, p_{95}] \to [0, 255]$, whitening background noise while preserving anti-aliased stroke edges.

3. **Interior-Crop 90°/180°/270° Orientation Detection:** Using an adaptive binary ink mask $M(x, y) = \mathbb{I}[I(x, y) < \frac{1}{2}(p_5 + p_{80})]$, we compare horizontal row-projection variance $\text{Var}_y(\mu_x(M))$ against vertical column-projection variance $\text{Var}_x(\mu_y(M))$. When $\text{Var}_x > 2.2\,\text{Var}_y$, the page is sideways. Crucially, to prevent dark scanner borders along the outer edges of ID card scans from biasing orientation, we crop to the interior $[15\%, 85\%]$ sub-window M_{int} and compare left-half vs. right-half ink density to select 90° CCW vs. 270° CCW rotation.

4. **Two-Stage Projection-Profile Deskew:** We sweep rotation angles $\theta \in [-12°, +12°]$ (coarse 1.0° step followed by fine 0.2° step) to maximize the horizontal row-sum objective:

$$\mathcal{J}(\theta) = \text{Var}(r_\theta) + \frac{1}{2}\mathbb{E}\left[(\Delta r_\theta)^2\right], \quad r_\theta(y) = \sum_x M_\theta(x, y) \tag{3}$$

   applying bicubic deskew when $|\theta^*| \geq 0.4°$ and $\mathcal{J}(\theta^*) > 1.03\,\mathcal{J}(0°)$.

5. **Receptive-Field Lanczos Resizing & Spectral Layout Reconstruction:** Images with short edge < 900 px or embedded DPI < 130 are upscaled via Lanczos interpolation, while oversized scans (> 2560 px) are downscaled. Batched ONNX `PP-OCRv6` (`oar-ocr-cli`) bounding boxes are grouped into horizontal text lines and ordered via `layout_fast.order_blocks` spectral layout routing.

2.3 Stage 3: Zero-Hallucination Span Extraction via Pydantic-Compiled GLiNER 2.5

Rather than prompting a generative LLM, Sovereign-OCR defines domain schemas as strongly-typed Pydantic v2 models (`InvoiceHeaderSchema`, `RemitToPayeeSchema`, `DocumentClassificationSchema`) with embedded regular-expression validators and confidence thresholds. Our compiler (`build_gliner2_schema_from_pydantic`) translates these models directly into: (i) `gliner2.Schema` structures for local `fastino/gliner2.5-multi-v1` (`BoundaryExtractor`) and (ii) flat entity labels for Rust ONNX `gliner_fast_py` (`gliner_multi_v2.1` FP32).

Because a span-boundary extractor scores start/end token pairs (s_i, e_i) over the input sequence T and extracts substrings $T[s_i : e_i]$, any output field f satisfies $f \sqsubseteq T$. Consequently, the probability of generating an out-of-document hallucinated digit string is identically zero. Extracted spans are further governed by deterministic algebraic validators:

- **Brazilian CPF (11-digit Mod-11) & CNPJ (14-digit Mod-11):** Verifies both cascading check digits d_{10}, d_{11} (CPF) and weight vectors w_1, w_2 (CNPJ), eliminating false matches on phone or account numbers.
- **Mobile Device IMEI (15-digit Luhn):** Enforces $\sum_{i=0}^{14} \text{luhn}(d_i, i) \equiv 0 \pmod{10}$ across TAC prefixes (`35*`, `86*`).

- **NF-e Chave de Acesso (44-digit Mod-11):** Verifies valid IBGE state UF prefixes (`11..53`) and the 43-digit weighted Mod-11 check digit d_{44} across whitespace-segmented DANFE headers.

2.4 Stage 4: Active Runtime Ontology Grounding

In enterprise AP workflows, text extraction alone is insufficient to produce a valid ERP posting payload. `SovereignOntologyEngine` loads the organization's OWL/Turtle knowledge graph and historical ledger distributions into an in-memory multi-index (exact name, street number + street token, postal ZIP5, phone/tax ID, and `align_fast` fuzzy token overlap). During extraction:

1. **Plant/Bill-To ZIP Exclusion:** ZIP codes belonging to the buyer's own facilities are masked during vendor address matching so "Ship To / Bill To" blocks are never misidentified as the supplier.

2. **Supplier (`LFA1`) vs. Permitted Payee (`LFB1`) Resolution:** When a remittance lockbox or factoring company is detected, the engine binds both the primary vendor ID and the permitted payee ID/address from the ontology.

3. **Historical Expense GL & Cost Center Governance:** When selecting default General Ledger (GL) accounts from historical vendor postings, SAP AP reconciliation accounts (`21100001`) are filtered out to guarantee selection of the true expense GL account, while Cost Centers and Profit Centers are filtered by the target Company Code prefix.

4. **Fiscal Tax State Machine (`I1` vs. `U1` vs. Exempt):** If positive sales tax is extracted on the invoice, tax code `I1` is assigned; if zero tax is billed on an `I1`-default vendor, the tax code is cleared to exempt (`""`); if the vendor is configured for Consumer Use Tax (`U1`) and bills $0.00 tax, `U1` is preserved for self-assessment.

2.5 Stage 5: Deterministic RDF 1.2 Provenance & Single-Binary Packaging

Every governed extraction bundle is compiled by `quad_fast` into an RDF 1.2 dataset containing five named graphs (`SystemManifest`, `OntologyAlignment`, `SourceArtifacts`, `ExtractedFacts`, and `GovernanceDecisions`). Quads are sorted canonically and serialized into a Zstd-compressed Apache Arrow IPC stream, yielding a deterministic 16-hex-character SHA-256 receipt hash that is 100% bit-identical across repeated executions.

For air-gapped enterprise deployment, all eight Rust PyO3 `abi3` extension modules (`email_fast`, `pdf_fast`, `pdf_fast_write`, `docx_fast`, `xlsx_fast`, `layout_fast`, `quad_fast`, `gliner_fast_py`) and the compiled `oar-ocr-cli` ONNX binary are packaged into a single 25.92 MB PEP-427 wheel (`cp38-abi3`) and a self-extracting PEP-441 standalone executable (`dist/sovereign-ocr`).

Table 1: Benchmark A (US Commercial SAP Accounts Payable, $N = 100$ POC invoices across 103 bundles / 138 files): Posting-field accuracy comparing the Legacy xSuite+LLM Consultant Baseline against **Sovereign-OCR**.

SAP Posting Field	Legacy LLM POC	Sovereign-OCR	Δ Gain
Document Ingested	69.0% (69/100)	**100.0**% (100/100)	+31.0%
SAP Doc Type (`KR/KG`)	0.0% (0/100)	**100.0**% (100/100)	+100.0%
SAP Vendor ID (`LIFNR`)	1.0% (1/100)	**100.0**% (100/100)	+99.0%
Vendor Name	58.0% (58/100)	**100.0**% (100/100)	+42.0%
Invoice Reference (`XBLNR`)	55.0% (55/100)	**100.0**% (100/100)	+45.0%
Invoice Amount (`WRBTR`)	59.0% (59/100)	**100.0**% (100/100)	+41.0%
Line Item Text (`SGTXT`)	0.0% (0/100)	**100.0**% (100/100)	+100.0%
Payment Terms (`ZTERM`)	0.0% (0/100)	**100.0**% (100/100)	+100.0%
Tax Code (`MWSKZ`)	43.0% (43/100)	**100.0**% (100/100)	+57.0%
GL / Cost / Profit Center	0.0% (0/100)	**100.0**% (100/100)	+100.0%
End-to-End Bundle Exact	0.0% (0/100)	**100.0**% (100/100)	+**100.0**%

Table 2: Ablation Study on Benchmark A ($N = 100$ SAP AP invoices), quantifying the exact contribution of each Sovereign-OCR stage to end-to-end bundle accuracy.

Pipeline Configuration	Ingest	Ref/Amt	Vendor/GL	Bundle Exact
1. Legacy xSuite + LLM POC	69.0%	55.0%	1.0%	0.0%
2. w/o Multi-Format (`.msg`/`.xlsx`)	69.0%	69.0%	69.0%	69.0%
3. w/o Self-Healing CMap/OCR Gate	100.0%	35.0%	35.0%	35.0%
4. w/o Active Ontology Grounding	100.0%	86.0%	0.0%	0.0%
5. w/o Payee (`LFB1`) & Tax Gov.	100.0%	100.0%	77.0%	61.0%
Full Sovereign-OCR (5 Stages)	**100.0**%	**100.0**%	**100.0**%	**100.0**%

3. Experimental Evaluation

We evaluate Sovereign-OCR on two production enterprise datasets spanning distinct languages (English and Portuguese), document topologies, and regulatory domains. All experiments run 100% offline on an Apple M-series workstation using the single 25.92 MB `sovereign-ocr` distribution.

3.1 Benchmark A: Enterprise SAP Accounts Payable

Benchmark A comprises 100 production SAP/xSuite Accounts Payable transactions (103 document bundles containing 138 individual files: 69 direct PDFs, 29 Outlook `.msg` containers with nested PDF/DOCX/image attachments, and 2 `.xlsx` workbooks) grounded against an enterprise ontology of 22,630 OWL/RDF triples (2,469 active vendors across multiple SAP company codes).

Table 1 compares Sovereign-OCR against the prior xSuite + commercial LLM consultant baseline. The legacy pipeline completely failed to unpack the 31 non-PDF containers (69.0% ingestion ceiling), achieved only 55.0% on invoice references and 59.0% on invoice amounts due to corrupted CMap tables and balance-forward confusion, and scored 1.0% on SAP Vendor IDs and 0.0% on GL/Cost Center coding. Sovereign-OCR achieves **100.0**% across all 10 SAP posting fields, executing the full 103-bundle benchmark in 3.71 s (36 ms/bundle with warm OCR cache; 150 ms/bundle cold).

Ablation Analysis (Table 2): Removing Stage 1 multi-format unpacking (`email_fast`, `xlsx_fast`, `docx_fast`) drops coverage by exactly 31.0% (29 `.msg` + 2 `.xlsx` bundles). Disabling the Stage 2 Self-Healing

Table 3: Benchmark B (*Assurant Documentoscopia* Brazilian Insurance Claims, 8 Claims, 88 Files: 44 Primary Documents + 44 Empty `.retries`): Ablation and baseline comparison across key claim verification tasks.

Pipeline Configuration	Role Cls. (88 files)	CPF (8/8)	IMEI (8/8)	NF-e Key (3/3)	BO Type (8/8)
1. Native PDF (`daase_current`)	58.0%	25.0%	12.5%	33.3%	12.5%
2. Prior OCR (`daase_ocrfix`)	90.9%	100.0%	100.0%	66.7%	100.0%
3. Sovereign w/o $\Delta = +29$ CMap Fix	100.0%	100.0%	100.0%	66.7%	100.0%
Full Sovereign-OCR	**100.0**%	**100.0**%	**100.0**%	**100.0**%	**100.0**%

Table 4: Sideways Brazilian Identity Card (`44087628R / Documento de identificação`, 740 × 1953 px): Impact of Stage 2 Interior-Crop 90° Rotation and DPI-Aware Lanczos Upscaling (1.2162× → 2375 × 900 px).

Metric / Extracted Field	Native PDF	Prior `ocrfix`	Sovereign-OCR
Detected Orientation	0° (None)	Unrotated	**90°** CCW (Interior)
Preprocessed Resolution	740 × 1953	740 × 1953	**2375 × 900** (1.22×)
Recovered Character Yield	0 chars	218 chars	**572** chars (**2.62**×)
RG Identifier (`02.724.762-77`)	Missed	Partial	**Exact Match**
Issuance Date (`08-10-2020`)	Missed	`06-10-2020`	**Exact** (`2020-10-08`)
Mother's Name (`... DE JESUS`)	Missed	`... DESUS`	**Exact** (`... DE JESUS`)
Birth Date (`05-11-1967`)	Missed	Exact	**Exact** (`1967-11-05`)
CPF (`481.922.605-34`, Mod-11)	Missed	Exact	**Exact** (`48192260534`)

CMap & Visual OCR Gate causes the largest accuracy drop (−65.0%, down to 35.0%), as 65/100 invoice bundles contain either scanned image pages or corrupted `ToUnicode` font tables. Removing Stage 4 Active Ontology Grounding leaves raw text extraction intact (86.0%) but drops SAP Vendor ID and Expense GL/Cost Center resolution to 0.0%. Disabling Permitted Payee (`LFB1`) and Tax State Machine governance degrades 23 factoring/lockbox bundles and 23 tax-governed bundles (61.0% bundle exact).

3.2 Benchmark B: Cross-Lingual Generalization to Brazilian Insurance Claims (*Documentoscopia*)

To test whether Sovereign-OCR generalizes without company-specific tuning to a completely different language and domain, we evaluate it on **Assurant Documentoscopia**: 8 Brazilian mobile theft/robbery insurance claims (88 files: 44 primary documents spanning *Aviso de Sinistro*, *Boletim de Ocorrência (BO)*, *Nota Fiscal / DANFE*, *Comprovante de Residência (CR)*, and *RG/CNH* identity documents, plus 44 zero-byte `.retries` artifacts).

As shown in Table 3, 81/88 files in the native PDF baseline (`daase_current`) fail due to scanned imagery or shifted Identity-H CMap tables, yielding only 2/8 CPFs and 1/8 IMEIs. Even after standard OCR (`daase_ocrfix`), 42/44 non-empty files retained shifted Type0/Identity-H tokens (e.g., `%2/(7,0`, `'HVFULomR`, `2SHUDGRUD`, `5HVLGrQFLD`) because the upstream PDF renderer painted the shifted CMap glyphs onto the rasterized canvas, and 1/3 of the 44-digit NF-e access keys failed extraction. With Sovereign-OCR's algebraic $\Delta = +29$ CMap repair and Mod-11/Luhn validators, role classification reaches **100.0**% (88/88 files), validated CPF and 15-digit Luhn IMEI extraction reach **100.0**% (8/8 claims), and 44-digit NF-e access key recovery increases from 66.7% to **100.0**% (3/3).

Table 4 highlights the impact of our 5-step visual preprocessor on the sideways Brazilian RG identity card in claim `44087628R` (740 × 1953 px). Because the scan has a dark black scanner border along the left edge, naive full-frame ink-density heuristics rotate the image 270° (upside-down). Sovereign-OCR's interior-crop [15%, 85%] variance test correctly selects 90° CCW rotation and upscales the short edge by 1.2162× to 2375 × 900 px, recovering 572 characters (2.62× more than `ocrfix`) and eliminating the date (`06-10-2020` → `08-10-2020`) and filiation (`DESUS` → `DE JESUS`) recognition errors.

4. Conclusion

We presented **Sovereign-OCR**, an air-gapped, single-binary neuro-symbolic document intelligence engine that unifies native Rust multi-format unpacking, algebraic PDF CMap self-healing, 5-step geometry/DPI-aware visual preprocessing, zero-hallucination Pydantic-to-`GLiNER 2.5` span extraction, active runtime ontology grounding, and deterministic RDF 1.2 provenance. Evaluated across 100 US SAP Accounts Payable bundles and 88 Brazilian insurance claim files, Sovereign-OCR achieves 100.0% accuracy at sub-second latency without external cloud dependencies.

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